# CV of the Researcher

## Personal details

**Name:** Muhammad Ahsan
**Data and place of birth:** September 19, 1990, Sahiwal, Pakistan
**Countr/Nationality:** Pakistan/Pakistani
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**Research unique identifier(s): ORCID:** 0000-0003-2362-3297
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**Research Interests:** Development of hybrid AI and physics-informed frameworks for condition monitoring and fault diagnosis of complex industrial systems, with a focus on explainable decision support that integrates machine learning, signal processing, and domain knowledge.

## Professional activities

- 01/10/2025 to 30/03/2026. Postdoctoral Researcher (Research Assistant Professor) at Faculty of Mechanical Engineering and Robotics, AGH University of Science and Technology in Krakow, Poland. **Research Topic:** Development of an integrated framework for SCADA-based condition monitoring and fault detection of wind turbines, incorporating image analysis and trend monitoring techniques.
- 01/10/2020 to 30/09/2024. Worked on my PhD project entitled "Vehicle Diagnostics using Artificial Intelligence and Digital Signal Processing Methods" where I utilized low-cost MEMS accelerometer to capture vibration data from the vehicle engine. The recorded data was analysed using advanced digital signal processing techniques and artificial intelligence models to accurately diagnose engine health conditions. The PhD project was carried out at the Silesian University of Technology, Poland
- 13/12/2024 to 23/12/2024. I visited Department of Networks and Digital Media, Kingston University London, London, UK as a visiting researcher
- 01/07/2022 to 31/10/2022. I visited Dynamics and Vibrations Group, Department of Science and Engineering Methods, University of Modena and Reggio Emilia, Riggio Emilia, Italy as a visiting researcher
- 13/09/2021 to 26/09/2021. I visited Department of Networked Systems and Services, Budapest University of Technology and Economics, Budapest, Hungary for research internship

## Academic Background

- 15/07/2025. Doctor of Philosophy in the field of engineering and technical science, and discipline of automation, electronics, electrical engineering and space technologies, Silesian University of Technology, Gliwice, Poland. (Doctoral dissertation defended on 09/07/2025 and resolution on award the degree of Doctor of Philosophy issued on 15/07/2025)
- 19/06/2020. Master of Control Theory and Control Engineering, Nanjing University of Science and Technology, Nanjing, China
- 24/04/2014. Bachelor of Electrical Engineering (Telecom), Air University, Islamabad, Pakistan

## Selected Publications

### A. Peer-Reviewed Journal Papers

[J.1] **M. Ahsan** & P.B. Dao. (2026). **Title:** Multi-Channel SCADA-Based Image-Driven Power Prediction for Wind Turbines using Optimized LeNet-5-LSTM Hybrid Neural Architecture. **Journal:** Energies, 19(5), 1-29. DOI: https://doi.org/10.3390/en19051169 , Q1, IF: 3.2

**[J.2]** **M. Ahsan**, J. Rodriguez & M. Abdelrahem. (2025). **Title:** Bearing Fault Diagnosis in Induction Motors using Low-Cost Triaxial ADXL335 Accelerometer and a Hybrid CWT-

DCNN-LSTM Model. **Journal:** IEEE Access, Vol. 13, pp. 101037 – 101050. DOI: https://doi.org/10.1109/ACCESS.2025.3577672 , Q1, IF: 3.6

**[J.3]** **M. Ahsan**, M.W. Hassan, J. Rodriguez & M. Abdelrahem. (2025). **Title:** Enhanced Fault Diagnosis in Rotating Machinery using a Hybrid CWT-LeNet-5-LSTM Model: Performance Across Various Load Conditions. **Journal:** IEEE Access, Vol. 13, pp. 1026 – 1045. DOI: https://doi.org/10.1109/ACCESS.2024.3522948 , Q1, IF: 3.6

**[J.4]** **M. Ahsan**, D. Bismor & P. Fabis. (2025). **Title:** Application of low-cost ADXL1002 accelerometer for vehicle engine misfire detection using a novel hybrid EMD-based image processing and DCNN-LSTM model. **Journal:** Vibrations in Physical Systems, Vol. 36(1), pp. 1 –7. DOI: https://doi.org/10.21008/j.0860-6897.2025.1.02 , Q4

**[J.5]** **M. Ahsan**, D. Bismor, & P. Fabis. (2024). **Title:** Analysis of the Vehicle Engine Misfires using Frequency-Domain Approaches at Various RPMs with ADXL1002 Accelerometer. **Journal:** Archives of Acoustics, Vol. 49(4), pp. 507 – 516. DOI: https://doi.org/10.24425/aoa.2024.148813 , Q3, IF: 0.9

**[J.6]** **M. Ahsan**, & M.M. Salah. (2023). **Title:** Similarity index of the STFT-based health diagnosis of variable speed rotating machines. **Journal:** Intelligent Systems with Applications, Vol. 20, pp: 1 – 9. DOI: https://doi.org/10.1016/j.iswa.2023.200270 , Q1, IF: 4.3

**[J.7]** **M. Ahsan**, & M.M. Salah. (2023). **Title:** Efficient DCNN-LSTM Model for Fault Diagnosis of Raw Vibration Signals: Applications to Variable Speed Rotating Machines and Diverse Fault Depths Datasets. **Journal:** Symmetry, Vol. 15(7), pp: 1 – 22. DOI: https://doi.org/10.3390/sym15071413 , Q1, IF: 2.2

**[J.8]** **M. Ahsan**, D. Bismor, & M.A. Manzoor. (2023). **Title:** ARL-Wavelet-BPF Optimization using PSO Algorithm for Bearing Fault Diagnosis. **Journal:** Archives of Control Sciences, Vol. 33(LXIX), pp: 589 – 606. DOI: https://doi.org/10.24425/acs.2023.146961 , Q3, IF: 1.2

**[J.9]** **M. Ahsan**, & D. Bismor. (2022). **Title:** Early-Stage Fault Diagnosis for Rotating Element Bearing using Improved Harmony Search Algorithm with Different Fitness Functions. **Journal:** IEEE Transactions on Instrumentation and Measurement, Vol. 71, pp: 1 – 9. DOI: https://doi.org/10.1109/TIM.2022.3192254 , Q1, IF: 5.9

## B. Peer-Reviewed Conference Papers

**[C.1]** M. Abdelrahem, **M. Ahsan** & J. Rodriguez. (September 2025). **Title:** Enhanced LeNet-5-LSTM-Based Diagnosis of PMSM Stator Faults using Vibration Signals Across Different Fault Severities. **Conference:** IEEE CPERE 2025, Pyramisa Island Aswan, Egypt, pp. 1 – 6. DOI: https://doi.org/10.1109/CPERE65146.2025.11240075

**[C.2]** **M. Ahsan**, D. Bismor & P. Fabis. (September 2024). **Title:** Comparison of ANN and CNN models for misfire detection in vehicle engine at different RPMs with low-cost ADXL1002 accelerometer. **Conference:** IEEE 2024 Signal Processing: Algorithms, Architectures, Arrangements, and Applications (SPA), Poznan, Poland, pp. 60 – 65. DOI: https://doi.org/10.23919/SPA61993.2024.10715607

**[C.3]** **M. Ahsan** & D. Bismor. (September 2023). **Title:** Advanced Fault Diagnosis in Rotating Machines using 2D Grayscale Images with Improved Deep Convolutional Neural Networks. **Conference:** IEEE 2023 Signal Processing: Algorithms, Architectures, Arrangements, and Applications (SPA), Poznan, Poland, pp. 77 – 82. DOI: https://doi.org/10.23919/SPA59660.2023.10274428

**[C.4]** **M. Ahsan** & D. Bismor. (June 2023). **Title:** Calibration of a High Sampling Frequency MEMS-Based Vibration Measurement System. **Conference:** XXI Polish Control Conference (PCC2023), Gliwice, Poland, Vol. 708, pp. 294 – 302. DOI: https://doi.org/10.1007/978-3-031-35170-9_28

**[C.5]** **M. Ahsan** & D. Bismor. (April 2022). **Title:** Early-Stage Faults Detection Using Harmony Search Algorithm and STFT-Based Spectral Kurtosis. **Conference:** Automation 2022: New Solutions and Technologies for Automation, Robotics and Measurement Techniques, Warsaw, Poland, Vol. 1427, pp. 75 – 84. DOI: https://doi.org/10.1007/978-3-031-03502-9_8

## Honours and Awards

- **16/02/2026: Seal of Excellence – HORIZON-MSCA-2025-PF-01-01**: Awarded Seal of Excellence for project proposal submitted under the Horizon Europe Marie Skłodowska-Curie Action call and was recognised as a high-quality project proposal in a highly competitive evaluation process (Project Code: 101284098—COBRAIN)
- **31/12/2023: Best PhD Student, under the Excellence Initiative – Research University Programme**: Funded by the Rector of the Silesian University of Technology, Gliwice, Poland (Project Code: RJO15.5033.450.2023)
- **01/10/2020 to 30/09/2024: Polish Government Scholarship**: Funded by the Polish Government for PhD study and research at the Silesian University of Technology, Gliwice, Poland (Student Number: 4949)
- **24/04/2014: Silver Medal** awarded by the Vice Chancellor of Air University, Islamabad, Pakistan on maintaining second highest CGPA in the Department of Electrical Engineering (BETE-Fall-2009)
- **01/03/2010 to 31/08/2013: Merit Based Scholarship** awarded by the Department of Electrical Engineering of Air University, Islamabad, Pakistan for attaining the CGPA higher than 3.5 out of 4.0 (090636)

## Grants & Research Projects

**21/12/2023: Pro-quality Competition for Rector's Grants for Highly Scored Publications**

- **Funded** by the Rector of the Silesian University of Technology, Gliwice, Poland (Project Code: 32/014/RGJ24/2021)

**26/06/2023: PCC2023 Conference Grant for Young Authors**

- **Funded** by XXI Polish Control Conference, Poland to attend and present research work

**01/07/2022 to 31/10/2022: Erasmus+ Mobility Grant**

- **Co-funded** by the European Union; for research on vibration-based condition monitoring of mechatronic systems using AI and DSP (Project Code: 2021-1-PL01-KA131-HED-000005374)

**13/09/2021 to 26/09/2021: NAWA PROM Mobility Grant**

- **Co-funded** by the European Social Fund; for research on artificial intelligence algorithms (Project Code: 08/030/FSD20/0078)

**01/10/2025 to 30/03/2026: IDUB Postdoctoral Research Project**

- **Research Topic:** Development of an integrated framework for SCADA-based condition monitoring and fault detection of wind turbines, incorporating image analysis and trend monitoring techniques
- **Funder:** AGH University of Krakow (Excellence Initiative – Research University, IDUB Project Code: 501.696.7996)
- **Outcome:** Publication in *Energies* (Q1, IF: 3.2)

**01/10/2020 to 30/09/2024: Doctoral Research Project**

- **Research Topic:** Vehicle Diagnostics using Artificial Intelligence and Digital Signal Processing Methods

- **Funder:** Polish Government Scholarship + Silesian University of Technology
- **Outcome:** Prepared Doctoral Dissertation and Defended on 07/2025

**01/06/2023 to 31/12/2023: Young Scientist Research Grant** (Mini Project)
- **Research Topic:** Condition monitoring and fault diagnosis using advanced machine learning methods
- **Funder:** Silesian University of Technology (Project Code: 02/050/BKM23/0036)
- **Outcome:** Led to peer-reviewed publications and presentations including:
  [1] XXI Student Scientific Session & LXX Open Seminar on Acoustics – OSA 2024 (Presentation)
  [2] IEEE SPA 2023 and 2024 Conference (Paper Presentation and Publication)
  [3] Vibrations in Physical Systems (Q4)

**01/06/2021 to 31/12/2021: Young Scientist Research Grant** (Mini Project)
- **Research Topic:** Filter design using evolutionary algorithm for estimation in application to non-stationary signal analysis
- **Funder:** Silesian University of Technology (Project Code: 02/050/BKM21/0021)
- **Outcome:** Led to peer-reviewed publications including:
  [1] IEEE trans. on Instrumentation and Measurement (Q1, IF: 5.9)
  [2] Archives of Control Sciences (Q3, IF: 1.2)
  [3] Automation 2022 (Book Chapter)

**01/09/2017 to 31/03/2020: Master Research Project**
- **Research Topic:** Containment Control Problem for Multi-Agent Systems
- **Funder:** Chinese Government Scholarship (CSC) + Nanjing University of Science and Technology (Project Code: CSC No. 2017SLJ018830)
- **Outcome:** Prepared Master Dissertation and Defended on 03/2020

## Teaching and Pedagogical Experience

**01/10/2020 to 30/09/2024: Teaching Assistant**
Department of Measurements and Control Systems, Silesian University of Technology, Poland
- I was responsible for delivering high-level laboratory instruction to engineering students. My teaching focuses philosophy on bridging theoretical mathematical concepts with practical, hands-on industrial applications. I conduct laboratory classes in the following core areas:
  (a) Numerical Methods
  (b) Fundamental of Measurements
  (c) Computer Networks
  (d) System Identification

**25/01/2022 to 03/03/2022: Pedagogical Training and Project-Based Learning (PBL)**
European University on Responsible Consumption and Production
- A cornerstone of my teaching development was the completion of a specialized course on the "Implementation of Interdisciplinary Project-Based Learning (PBL) Projects", offered by the European University on Responsible Consumption and Production. Through this training, I mastered the "Student-Centered" approach, moving beyond traditional rote learning to facilitate collaborative, problem-solving environments.

## Professional Courses

- 30/04/2025. **Introduction to Guided Wave Testing V1.0**: Successfully completed training course (90 minutes) administered by Eddyfi Technologies
- 19/04/2022 to 21/04/2022. **Approximation of mass and damping matrices in structural dynamics**: Conducted by Prof. Robert Nemeth, Faculty of Structural Mechanics, Budapest University of Technology and Economics, Hungary

- 19/07/2021 to 05/08/2021. **International Virtual Course on Multidisciplinary Optimization and Machine Learning for Engineering Design Applications**: Organized by ITB (Indonesia) – Tohoku University (Japan) – HKUST (Hong Kong)
- 01/04/2021 to 08/04/2021. **Reinforcement Learning Virtual School**: Organized by Artificial and Natural Intelligence Toulouse Institute (ANITI)
- 15/04/2021 to 15/07/2021. **Responsible Production & Consumption**: Organized by the European University on Responsible Consumption and Production
- Attended following continuing professional development (CPD) courses organized by the Pakistan Engineering Council:

[1] 23/12/2025. Data Warehousing for Smart Infrastructure, Energy & Utilities
[2] 25/09/2025. Enhancing Engineers' Communication & Presentation Skills
[3] 16/09/2025. Artificial Intelligence in Engineering Applications, Innovations & Future Trends
[4] 31/08/2025. Risk Assessment in Engineering Industry
[5] 16/08/2025. Project Team Building in Engineering Project
[6] 09/08/2025. Health, Safety, and Environment (HSE) Management in Construction Industry
[7] 13/05/2025. Service Leadership and Professional Ethics
[8] 30/04/2025. Mastering Digital Tools for Engineering Works
[9] 01/03/2025. Potential of Hydrogen as Energy Source for Industry in Pakistan

## Other relevant information

### A. Languages

- Candidate speaks and writes English fluently (entire education in English as medium of instruction)

### B. Professional Memberships

- Registered Engineer with Pakistan Engineering Council (PEC), Islamabad, Pakistan (Reg. No. ELECT/39269)
- Member of the International Association of Engineers (IAENG) (Member No. 141064)

### C. Peer-Review Activities

- Reviewer for IEEE Access, IEEE Transactions, and MDPI journals

### D. Conference Presentations (Abstracts)

- **M. Ahsan**, D. Bismor. (Mar. 2022). **Title:** Application of artificial intelligence methods to signal enhancement for predictive maintenance of rotating element bearings. **Conference:** 50th Jubilee Winter School on Wave and Quantum Acoustics, XLIX-th Winter School on Environmental Acoustics and Vibroacoustics, Szczyrk, Poland. **DOI:** http://doi.org/10.24425/aoa.2022.141658
- **M. Ahsan** (Jun. 2023). **Title:** Sustainable Solutions for Condition Monitoring and Fault Diagnosis in Industrial Mechatronics Systems: A Focus on Vibration Signals-based Monitoring with Deep Learning. **Conference:** XXI Student Scientific Session, Poland.
- **M. Ahsan**, D. Bismor, & P. Fabis. (Sep. 2024). **Title:** A Novel Approach to Vehicle Engine Misfire Detection with ADXL1002 Accelerometer using EMD-Based Image Processing and DCNN-LSTM Model. **Conference:** LXX Open Seminar on Acoustics – OSA 2024, Poland. URL: https://osa2024.ippt.pan.pl/ksiazka-abstraktow

### E. Doctoral Dissertation

- **M. Ahsan** (Jul. 2025). **Title:** Vehicle diagnostics using Artificial Intelligence and digital signal processing methods. **Institute:** Silesian University of technology, Gliwice, Poland. **URL:** https://repolis.bg.polsl.pl/dlibra/publication/90344/edition/80858/content